

Label-Invariant Hessian Regularization Mitigates Grokking in Mathematical Reasoning

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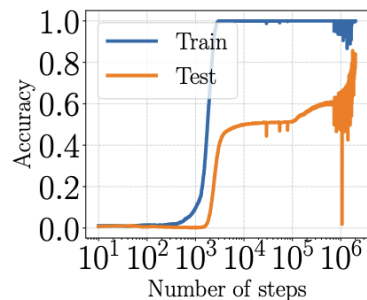
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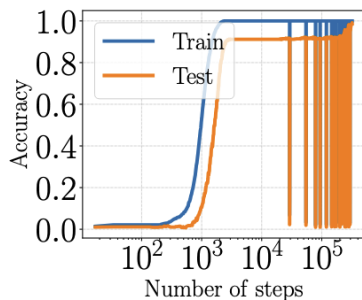
Grokking phenomenon

What is Grokking?

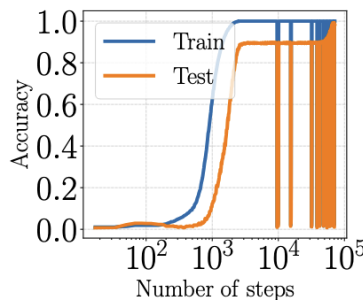
- Grokking is a delayed generalization phenomenon in modular arithmetic tasks, such as $(a + b) \bmod p$.
- A model (e.g., a transformer) first achieves 100% training accuracy, while test accuracy remains low.
- After many more training epochs, test accuracy increases.



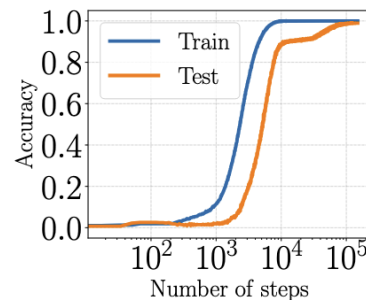
(a) SGD, $\lambda = 0$



(b) SGD, $\lambda = 0.1$



(c) SGD, $\lambda = 1$



(d) Noise injection, $\lambda = 1$



Our approach: Investigating the Hessian

Problem: Grokking is slow and unstable.

Our first insight: We regularize the Hessian of loss.

Implementation: Let f_W denote a transformer model and let $\ell(f_W(x), y)$ denote the loss function of one sample x, y . Consider the following noise injection with a Gaussian noise U :

$$W \leftarrow W - \eta \nabla_W \ell(f_{W+U}(x), y),$$

Then the expected loss is approximately:

$$\ell(f_W(x), y) + \frac{\sigma^2}{2} \text{Tr}[\nabla^2 \ell(f_W(x), y)] + O(\sigma^3)$$



Our approach: Investigating the Hessian

Results: Hessian regularization stabilizes the training curves and reduces the number of grokking steps.

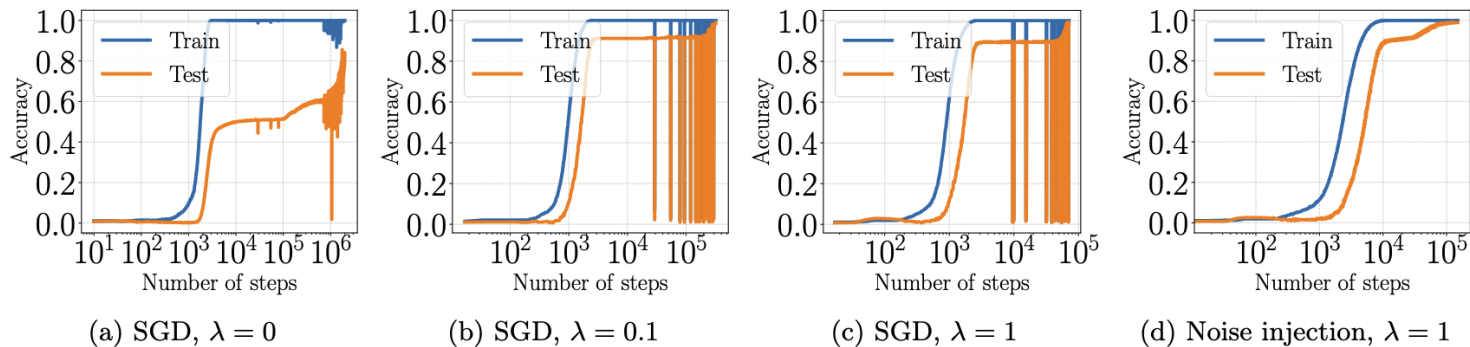


Figure 1: Comparing the training curves of transformer models for modular arithmetic tasks using weight decay (with varying levels indicated by λ), and Hessian regularization (via noise injection).



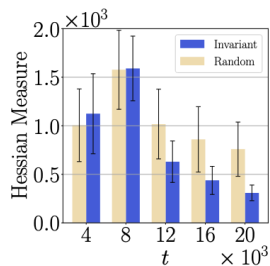
Label-Invariance in the Loss Landscape

Question: Why does the Hessian matter? We investigated the structure of the loss surface during training.

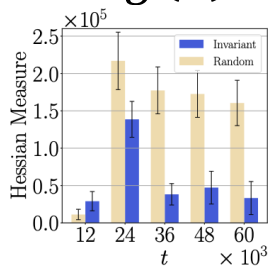
Key Finding: The loss surface becomes exceptionally **flat** (low Hessian values) specifically along **label-invariant** directions.

Invariant directions:

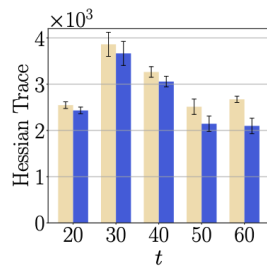
- For an input $x = (a, b)$ in $(a + b) \bmod p$.
- A transformation $g(x) = (a + k, b - k) \pmod{p}$.
- An invariant direction is $v = g(x) - x$



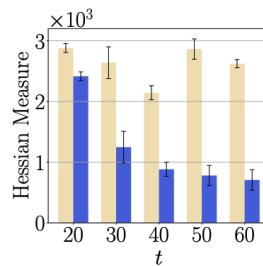
(a) Modulo addition



(b) Modulo quadratic



(c) Shortest distance



(d) Triangle counting

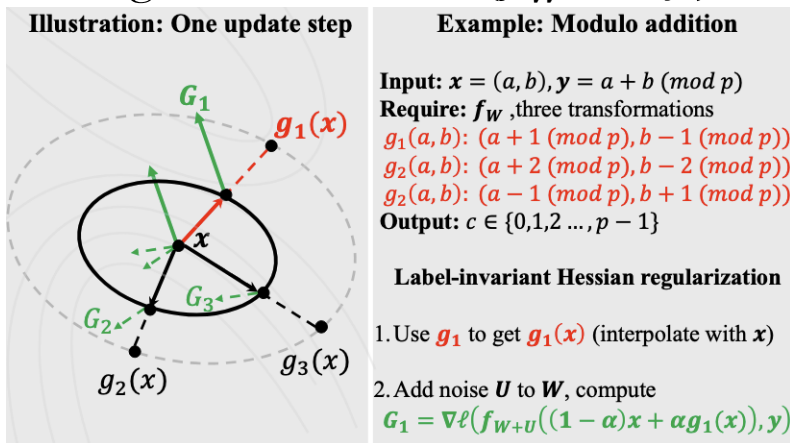


Label-Invariant Hessian Regularization (LIHR)

Idea: Explicitly enforce the flatness along these invariant directions.

Algorithm:

1. **Sample Invariance:** Pick a random invariant transformation $g \sim \mathcal{G}$.
2. **Interpolate:** Create an augmented sample $\tilde{x} = (1 - \alpha)x + \alpha g(x)$.
3. **Regularize Hessian:** Add noise to weights $\tilde{W} = W + U$.
4. **Update:** Compute the gradient $G = \nabla \ell(f_{\tilde{W}}(\tilde{x}), y)$.



Theoretical guarantee

The algorithm is supported by a PAC-Bayes generalization bound.

Assume that the Hessian is bound from a fixed constant

$$\alpha_{\mathcal{G}} := \sup_{w \in \mathcal{W}, g \in \mathcal{G}} w^{\top} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[[\nabla^2 \ell(f_W(g(x)), y)]^+ \right] w$$

Theorem 1

Suppose $\ell(f_W(\cdot), \cdot)$ is twice-differentiable in W and the Hessian $\nabla^2 \ell(f_W(\cdot), \cdot)$ is Lipschitz continuous. For any $W \in \mathcal{W}$, and small enough $\epsilon > 0$, and any small $\delta > 0$, with probability at least $1 - \delta$ over the randomness of n samples, we have:

$$L_{\mathcal{G}}(f_W) \leq (1 + \epsilon) \hat{L}_{\mathcal{G}}(f_W) + (1 + \epsilon) \sqrt{\frac{C \alpha_{\mathcal{G}}}{n}} + \xi$$

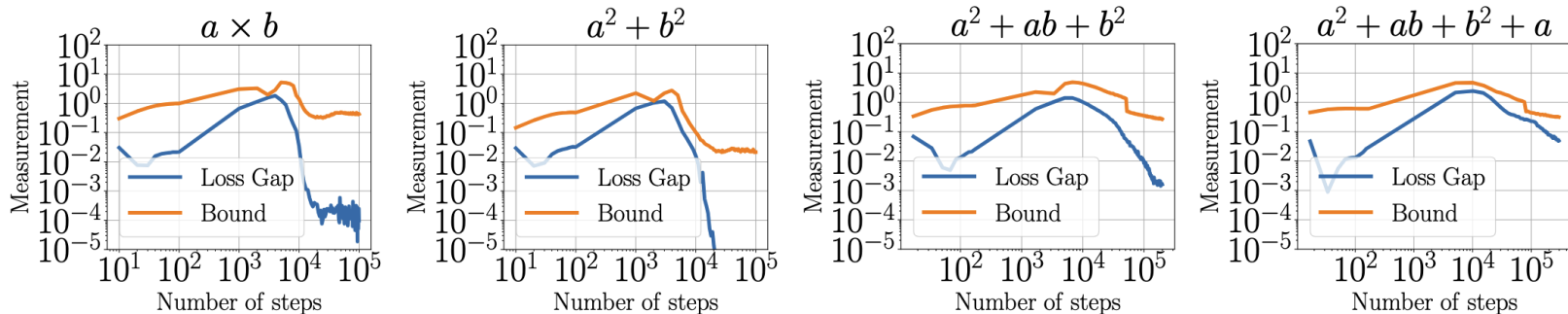


Generalization bound

- Based on **Theorem 1**, we derive a bound on the generalization gap that depends on a Hessian-based measure α_G over the invariant transformations:

$$L_G - \hat{L}_G \lesssim \sqrt{\frac{C \alpha_G}{n}}$$

- This bound is shown to be non-vacuous when measured on transformers.



The Scaffolding: PAC-Bayes Bound

The proof starts with a standard linear PAC-Bayes bound.

This theorem formally connects the perturbed true error $L_Q(W)$ to the perturbed training error $\hat{L}_Q(W)$ with probability $1 - \delta$:

$$L_Q(W) \leq \frac{1}{\beta} \hat{L}_Q(W) + \frac{C \left(KL(Q \parallel \mathcal{P}) + \log \frac{1}{\delta} \right)}{2\beta(1 - \beta)n}$$

- $Q = \mathcal{N}(W, \Sigma)$ is a Gaussian posterior around W ; $\mathcal{P}$ is a Gaussian prior; $\beta \in (0,1)$, n = sample size.



The Bridge: Taylor Expansion of Perturbed Loss

We use a Taylor expansion to build the bridge between the perturbed loss ℓ_Q and the original loss ℓ . The expansion introduces the Hessian:

$$\ell_Q(f_W(x), y) - \ell(f_W(x), y) \leq \langle \Sigma, \nabla^2 \ell(f_W(x), y) \rangle + C_1 \|\Sigma\|_F^{\frac{3}{2}}$$

where Σ is the population covariance matrix of the perturbation, ∇^2 is the Hessian matrix of the loss function with respect to the weights of f_W .



Combining Hessian and KL terms

We then substitute the Hessian term and the KL divergence for Gaussian posteriors into the PAC-Bayes bound.

Next, consider the sum of the positive truncated Hessian term and the KL divergence term:

$$\langle \Sigma, [\nabla^2 \ell(f_W(x), y)]^+ \rangle + \frac{1}{n} \langle \Sigma^{-1}, vv^\top \rangle$$

The next step is to minimize this sum over the noise covariance:

$$\alpha_{\mathcal{D}} := \sup_{w \in \mathcal{W}} \sup_{(x,y) \sim \mathcal{D}} \frac{\sqrt{W^\top [\nabla^2]^+ W}}{\sqrt{n}}$$

To finish the proof of this theorem, we generalize the argument to the entire group-transformed data distribution under $\mathcal{G}$.



Generalization bound

By calculating the generalization gap, we have the final result:

$$\begin{aligned} & L_{\mathcal{G}}(f_W) - \hat{L}_{\mathcal{G}}(f_W) \\ &= \left(L_{\mathcal{G}}(f_W) - L_{\mathcal{Q},\mathcal{G}}(f_W) \right) + \left(L_{\mathcal{Q},\mathcal{G}}(f_W) - \frac{1}{\beta} \hat{L}_{\mathcal{Q},\mathcal{G}}(f_W) \right) \\ &+ \left(\frac{1}{\beta} \hat{L}_{\mathcal{Q},\mathcal{G}}(f_W) - \hat{L}_{\mathcal{G}}(f_W) \right) \end{aligned}$$

Final bound:

$$L_{\mathcal{G}}(f_W) \leq (1 + \epsilon) \hat{L}_{\mathcal{G}}(f_W) + (1 + \epsilon) \sqrt{\frac{C \alpha_{\mathcal{G}}}{n}} + \xi$$



Experiments

Setup:

- Tasks:
 - Modular arithmetic: 6 tasks:
$$a + b, a - b, a * b, \frac{a}{b}, a^2 + b^2 + ab, a^2 + b^2 + ab + a$$
 - Graph-algorithmic: node counting, shortest distance, and triangle counting.
- Models: 2-layer Transformers and small LLMs (Llama3-1B and Qwen3-8B).
- Baselines: Weight Decay, SAM, NSO, Grok Transfer .



Result 1: Mitigating Grokking in Transformers

Metric: Number of epochs to reach 100% test accuracy (in multiplies of 10^3).

Finding: On six modular arithmetic tasks, our algorithm **reduces the number of grokking steps by over 30%** on average compared to baseline methods.

Arithmetic Formula (mod $p = 97$)	$a + b$	$a \times b$	$a^2 + ab + b^2$	$a - b$	a/b	$a^2 + ab + b^2 + a$
Weight Decay	15.9 $\pm$ 1.1	15.7 $\pm$ 1.1	45.0 $\pm$ 2.2	25.3 $\pm$ 3.1	16.7 $\pm$ 2.5	48.6 $\pm$ 5.2
Dropout	15.1 $\pm$ 1.3	15.5 $\pm$ 1.5	40.6 $\pm$ 2.4	16.1 $\pm$ 1.1	15.3 $\pm$ 1.3	46.7 $\pm$ 7.1
SAM	16.3 $\pm$ 2.1	15.4 $\pm$ 1.2	88.7 $\pm$ 2.8	21.3 $\pm$ 1.3	21.5 $\pm$ 2.5	113.5 $\pm$ 26.1
Frob-SAM	15.2 $\pm$ 2.2	14.6 $\pm$ 2.5	46.2 $\pm$ 4.5	19.9 $\pm$ 1.2	20.4 $\pm$ 2.7	82.4 $\pm$ 2.4
NSO	8.3 $\pm$ 0.3	9.5 $\pm$ 0.1	54.5 $\pm$ 4.1	13.1 $\pm$ 1.0	15.1 $\pm$ 1.5	77.4 $\pm$ 3.5
Data Augmentation	14.0 $\pm$ 1.0	14.3 $\pm$ 1.1	53.2 $\pm$ 9.8	24.2 $\pm$ 3.1	22.4 $\pm$ 3.1	123.5 $\pm$ 11.8
GrokTransfer	6.9 $\pm$ 0.0	9.0 $\pm$ 0.1	68.9 $\pm$ 1.3	12.3 $\pm$ 2.0	13.2 $\pm$ 2.4	58.3 $\pm$ 3.1
LIHR-MAX	5.3$\pm$0.5	5.3$\pm$0.7	30.1 $\pm$ 2.6	14.5 $\pm$ 5.0	13.2 $\pm$ 2.1	76.6 $\pm$ 4.5
LIHR-RAND	6.7 $\pm$ 0.1	5.9 $\pm$ 0.1	29.3$\pm$2.4	12.1$\pm$0.9	9.2$\pm$0.1	47.2$\pm$2.2



Result 2: Generalization in LLMs

Metric: Test Accuracy.

Finding: Our algorithm **improves test accuracy by relevant 44%** on average over existing methods.

Text Inputs	$a + b$	$a \times b$	$a^2 + ab + b^2$	$a - b$	$\frac{a}{b}$	$a^2 + ab + b^2 + a$
Weight Decay	74.0 $\pm$ 0.1	74.1 $\pm$ 0.9	72.8 $\pm$ 0.1	49.9 $\pm$ 0.1	51.0 $\pm$ 0.1	50.1 $\pm$ 0.1
SAM	75.0 $\pm$ 0.1	74.5 $\pm$ 0.1	73.5 $\pm$ 0.0	49.9 $\pm$ 0.1	51.0 $\pm$ 0.1	50.1 $\pm$ 0.4
NSO	74.3 $\pm$ 0.1	74.8 $\pm$ 0.1	73.9 $\pm$ 0.1	49.9 $\pm$ 0.1	51.0 $\pm$ 0.1	50.1 $\pm$ 0.1
Data Augmentation	98.8 $\pm$ 0.1	99.7 $\pm$ 0.1	99.7 $\pm$ 0.1	99.9 $\pm$ 0.1	99.9 $\pm$ 0.1	85.5 $\pm$ 0.7
LIHR-MAX (2)	100.0 $\pm$ 0.0	99.7 $\pm$ 0.1	99.7 $\pm$ 0.1	99.9 $\pm$ 0.1	99.9 $\pm$ 0.1	78.6 $\pm$ 0.7
LIHR-RAND (1)	100.0 $\pm$ 0.0	99.7 $\pm$ 0.1	99.7 $\pm$ 0.2	99.2 $\pm$ 0.1	99.9 $\pm$ 0.1	90.5 $\pm$ 0.3

	Triangle Counting	Shortest Dist	Node Degree
ℓ_2	89.2 $\pm$ 0.4	89.1 $\pm$ 0.5	86.2 $\pm$ 0.4
SAM	90.3 $\pm$ 0.8	89.2 $\pm$ 0.3	85.7 $\pm$ 0.4
NSO	89.6 $\pm$ 0.3	89.2 $\pm$ 0.2	86.5 $\pm$ 0.2
Ours	95.8 $\pm$ 0.3	93.0 $\pm$ 0.2	99.1 $\pm$ 0.1



Result 3: Generalization in OOD settings

Setup: Train on numbers in $[0, p)$ and test on $[p, 2p - 1]$.

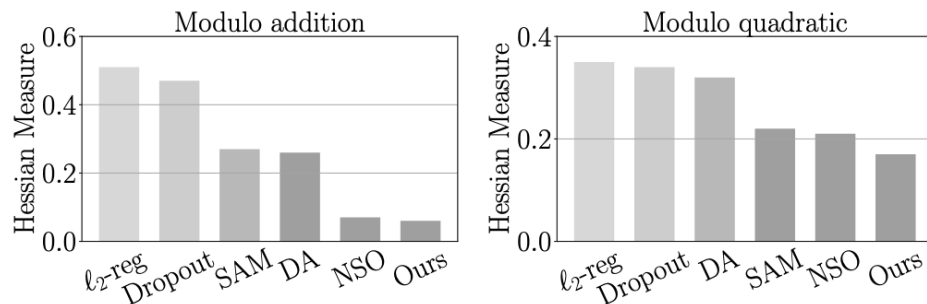
Finding: All baseline methods drop to less than 1% test accuracy, while our approach maintains 72% average accuracy.

Arithmetic formula (mod p)	$a + b$	$a \times b$	$a^2 + ab + b^2$
Weight decay	$\leq 1.0_{\pm 0.1}$	$\leq 1.0_{\pm 0.1}$	$\leq 1.0_{\pm 0.1}$
SAM	$\leq 1.0_{\pm 0.1}$	$\leq 1.0_{\pm 0.1}$	$\leq 1.0_{\pm 0.1}$
NSO	$\leq 1.0_{\pm 0.1}$	$\leq 1.0_{\pm 0.1}$	$\leq 1.0_{\pm 0.1}$
LIHR-RAND (Algorithm 1)	88.2 ± 0.3	92.6 ± 0.3	36.7 ± 0.7



Ablation

- Our approach reduces the Hessian measure.



- Both invariant data augmentation and Hessian regularization are essential. Using invariant data augmentation alone does not consistently reduce the required generalization steps. Noise injection reduces the generalization steps by an average of 30% compared to data augmentation. Our approach also improves over noise injection by 30%.



Extension: Hessian Spectrum

Motivation:

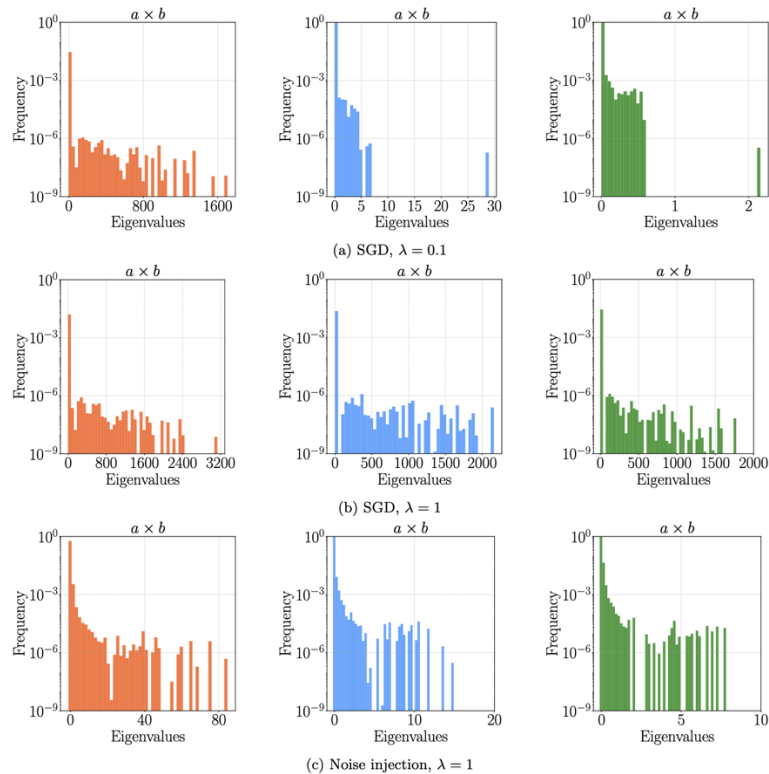
- The Hessian matrix contains far richer information than just its trace or largest eigenvalue.
- Analyzing the full Hessian spectrum provides a new lens for understanding the training dynamics of Transformers and LLMs.

Preliminary Findings:

- We approximate the Hessian spectrum at different training stages.
- We found that different regularization methods lead to distinct spectrum.
 - Strong Weight Decay ($\lambda = 1$): The model generalizes but converges to a sharp minimum, characterized by large Hessian eigenvalues.
 - Noise Injection (Hessian Reg.): The model converges to a flat minimum, evidenced by a spectrum with significantly smaller eigenvalues.



Extension: Hessian Spectrum



Extension: Hessian Spectrum

Problem: Grokking (delayed generalization) is a slow and unstable process in transformers.

Insights:

- **Hessian regularization** (via noise injection) **stabilizes** training and **reduces grokking steps** where standard weight decay introduces oscillations.
- The loss surface isn't randomly flat. It becomes specifically flat along **label-invariant input directions** as the model generalizes.

Our Solution: We proposed a new algorithm that combines Hessian regularization with label-invariant augmentation to explicitly enforce this flatness.

Thanks for listening!

